```
      DBName: discourse                                          ⑦
      Engine: postgres                                           ⑧
      EngineVersion: '12.10'                                     ⑨
      MasterUsername: discourse                                  ⑩
      MasterUserPassword: discourse
      VPCSecurityGroups:
      - !Sub ${DatabaseSecurityGroup.GroupId}
      DBSubnetGroupName: !Ref DatabaseSubnetGroup
    DependsOn: VPCGatewayAttachment
```

① Traffic to/from the RDS instance is protected by a security group.

② PostgreSQL runs on port 5432 by default.

③ The InstanceSecurityGroup resource is not yet specified; you'll add this later when you define the EC2 instance that runs the web server.

④ RDS also uses a subnet group to reference the VPC subnets.

⑤ The database resource

⑥ Disables backups; you want to turn this on (value > 0) in production.

⑦ RDS created a database for you in PostgreSQL.

⑧ Discourse requires PostgreSQL.

⑨ We recommend always specifying the version of the engine to avoid future incompatibility issues.

⑩ PostgreSQL admin username

PostgreSQL admin password; you want to change this in production.

Have you noticed the similarity between RDS and ElastiCache? The concepts are similar, which makes it easier for you to work with both services. Only one component is missing: the EC2 instance that runs the web server.

11.4.4 Virtual machine: Security group, EC2 instance

Discourse is a Ruby on Rails application, so you need an EC2 instance to host the application. Because it is very tricky to install the required Ruby environment with all its dependencies, we are using the recommended way to run Discourse, which is running a container with Docker on the virtual machine. However, running containers is not in scope of this chapter. You will learn more about deploying containers on AWS in chapter 18.

Discourse requires SMTP to send email

The forum software application Discourse requires access to an SMTP server to send email. Unfortunately, operating an SMTP server on your own is almost impossible, because most providers will not accept emails or flag them as spam. That's why we are using the Simple Email Service (SES) provided by AWS.